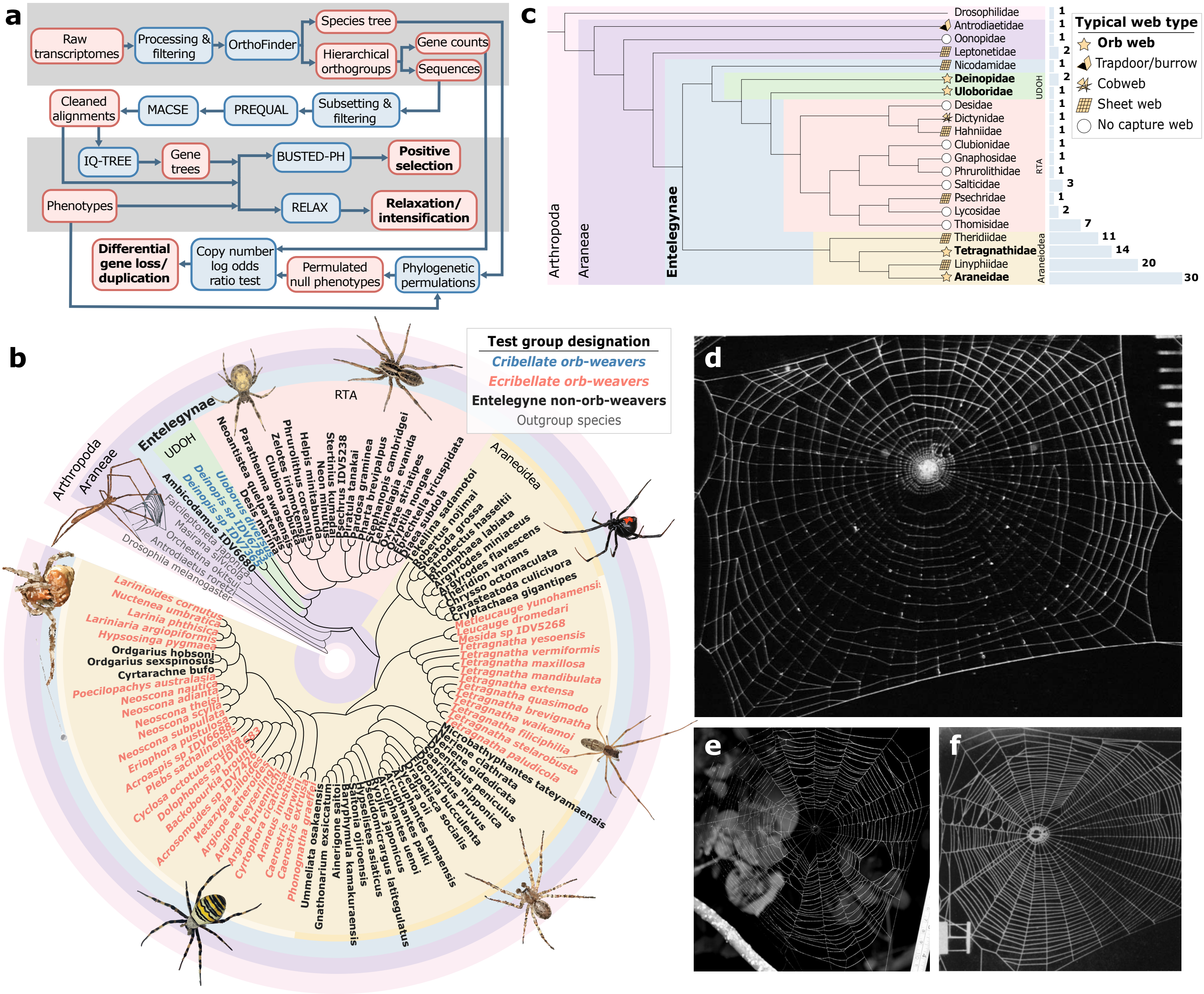
a
Raw transcriptomes
Processing & filtering
OrthoFinder
Species tree
Hierarchical orthogroups
Gene counts
Sequences
Cleaned alignments
MACSE
PREQUAL
Subsetting & filtering
IQ-TREE
Gene trees
BUSTED-PH
Positive selection
Phenotypes
RELAX
Relaxation/ intensification
Differential gene loss/ duplication
Copy number log odds ratio test
Permulated null phenotypes
Phylogenetic permutations
b
Test group designation
Cribellate orb-weavers
Ecribellate orb-weavers
Entelegyne non-orb-weavers
Outgroup species
Arthropoda
Araneae
Entelegynae
UDOH
RTA
Araneoidea
Drosophila melanogaster
Antrodiaetus roretzi
Orchestina okitsui
Masirana silvicola
Falcileptoneta japonica
Ambicodamus IDV6680
Deinopis sp IDV7365
Deinopis sp IDV6783
Uloborus diversus
Desis marina
Neoantistea quelpartensis
Paratheuma awasensis
Clubiona robusta
Zelotes iriomotensis
Phrurolithus coreanus
Helpis minitabunda
Neon minutus
Stertinius kumadai
Psechrus IDV5238
Piratula tanakai
Pardosa graminea
Pharta brevipalpus
Stephanopis cambridgei
Lehtinelagia evanida
Oxytate striatipes
Ozyptila nongae
Ebrechtella tricuspidata
Diaea subdola
Tekellina sadamotoi
Robertus nojimai
Steatoda grossa
Latrodectus hasseltii
Rhomphaea labiata
Argyrodes miniaceus
Argyrodes flavescens
Theridion varians
Chrysso octomaculata
Parasteatoda culicivora
Cryptachaea gigantipes
Metleucauge yunohamensis
Leucauge dromedari
Mesida sp IDV5268
Tetragnatha yesoensis
Tetragnatha vermiformis
Tetragnatha maxillosa
Tetragnatha mandibulata
Tetragnatha extensa
Tetragnatha quasimodo
Tetragnatha brevignatha
Tetragnatha waikamoi
Tetragnatha filiciphilia
Tetragnatha stelarobusta
Tetragnatha paludicola
Microbathyphantes tateyamaensis
Neriene clathrata
Neriene oidedicata
Doenitzius peniculus
Saaristoa nipponica
Doenitzius pruvus
Floronia bucculenta
Drapetisca socialis
Syedra oii
Arcuphantes tamaensis
Arcuphantes paiki
Arcuphantes uenoi
Ryojius japonicus
Pseudomicrargus latitegulatus
Hypselistes asiaticus
Saitonia ojiroensis
Baryphymula kamakuraensis
Ainerigone saitoi
Gnathonarium exsiccatum
Ummeliata osakaensis
Phonognatha graeffei
Caerostris extrusa
Caerostris darwini
Araneus inustus
Cyrtophora cicatrosa
Argiope bruennichi
Argiope keyserlingi
Argiope aetheroides
Metazygia zilloides
Acrosomoides sp IDV7426
Backobourkia brouni
Dolophones sp IDV6683
Cyclosa octotuberculata
Plebs sachalinensis
Acroaspis sp IDV6688
Eriophora pustulosa
Neoscona subpullata
Neoscona scylla
Neoscona theisi
Neoscona adianta
Neoscona nautica
Poecilopachys australasia
Cyrtarachne bufo
Ordgarius sexspinosus
Ordgarius hobsoni
Hypsosinga pygmaea
Lariniaria argiopiformis
Larinia phthisica
Nuctenea umbratica
Larinioides cornutus
c
Drosophilidae 1
Antrodiaetidae 1
Oonopidae 1
Leptonetidae 2
Nicodamidae 1
Deinopidae 2
Uloboridae 1
Desidae 1
Dictynidae 1
Hahniidae 1
Clubionidae 1
Gnaphosidae 1
Phrurolithidae 1
Salticidae 3
Psechridae 1
Lycosidae 2
Thomisidae 7
Theridiidae 11
Tetragnathidae 14
Linyphiidae 20
Araneidae 30
Arthropoda
Araneae
Entelegynae
UDOH
RTA
Araneoidea
Typical web type
☆ Orb web
Trapdoor/burrow
Cobweb
▦ Sheet web
○ No capture web
d
e
f